

exactly 80 mm: _____

START SOLVE

New! Here's Do the walkthrough once at cubeopath-3x.vercel.app/learn. This card is the memory jogger.

NOTATION

R = Right = turn that face clockwise, looking at it from outside the cube.
 F = Front = turn that face clockwise, looking at it from outside the cube.
 C = counter-clockwise = 2 = half turn. Lowercase r = that face plus the layer behind it turning together.
 y = spin the whole cube left, white staying down. A whole-cube turn solves nothing.

WORDS

algorithm = a fixed sequence of turns + trigger = a short algorithm your hands run as one unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – just work it out. Can't see it? Build the four white edges on the **YELLOW** face – first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
R'UR'



white sticker FRONT
L'UL



white sticker UP run the first one once to knock it out, then look again
R'UR'

3 MIDDLE EDGES

Top edge with XY yellow: Turn U until its front color matches the center under it. The arrow shows which slot it goes to.



goes RIGHT
U' R'UR' y L'UL



goes LEFT
U' L'UL U' y' R'UR'

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

R = R'UR' sexy move
 R'UR = R'UR' sane
 R'FRF' = R'FRF' sleighthammer
 gap = change grip

START SOLVE

New here? Do the walkthrough once at cubepath-six.vercel.app/

new to the memory jogger.

NOTATION

R = 180° = turn that face clockwise, looking at it from outside the cube.
 R' = 90° = turn that face counter-clockwise, looking at it from outside the cube.
 r = counter-clockwise 2 = half turn. Lowercase r f = that face plus the layer behind it, turning together.
 y = spin the whole cube left, white staying down. A whole-cube turn solves nothing.

WORDS

algorithm = a fixed sequence of turns · trigger = a short algorithm your hands run as one unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm = work it out. Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT

R'U'R'



white sticker FRONT

L'U'L



white sticker UP run the first one above to knock it out, then look again

R'U'R'

3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to.

goes RIGHT

U'R'U'

y

L'U'L

goes LEFT

U' L'U'L

y'

R'U'R'

Edge stuck in the wrong spot? Run either one to kick it out, then place it.

R
R'
U
U'
R
R'
F
F'
S
S'
R
R'
F
F'
S
S'
R
R'
F
F'
S
S'
R
R'
F
F'
S
S'
R
R'
F
F'
S
S'
R
R'
F
F'
S
S'
R
R'
F
F'
S
S'
R
R'
F
F'
S
S'
R
R'
F
F'
S
S'
R
R'
F
F'
S
S'
R
R'
F
F'
S
S'
R
R'
F
F'
S
S'
R
R'
F
F'
S
S'
R
R'
F
F'
S

TWO-LOOK

OLL CROSS -yellow edges



Line bar left-to-right
fRUR'U'f'



Hook hook pointing front-right
fRUR'U'f'

OLL CORNERS -yellow face



Sune $\frac{1}{2}$ yellow, rest clockwise
RUR'U RU2R'



Anti-Sune $\frac{1}{2}$ yellow, rest counter-clockwise
RU2R' U'RUR'



Pi $\frac{1}{2}$ yellow, heads left
fRUR'U'f' FRUR'U'f'

OLL CORNERS

continued



Headlights $\frac{1}{2}$ yellow, headlights at the back
R2R'U2 RD'R'U2R'



2x Headlights $\frac{1}{2}$ yellow, headlights both sides
RUR'U RU'R' URU2R'



Chameleon $\frac{1}{2}$ yellow next to each other
fUR'U'r' FRF'



Bowtie $\frac{1}{2}$ yellow diagonal
fR'UR'U'r' FR

FALLBACK

The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked; there is the worst.

Top not all yellow and nothing matches? Run Sun and read again.

No yellow edge at all? Run the first one, then read again.

■ **RUR'U** easy move ■ **RUR'U** easy move ■ **R'FRU'** sledgehammer gap = change grip

TWO-LOOK		OLL CORNERS <i>continued</i>	
	Line bar left-to-right F R U R U F		Headlights $\frac{3}{2}$ 2 yellow; headstitch at the back R 2 D 2 U 2 R D 2 U 2 R
	Hook hook pointing front-right f R U R U f		2x Headlights $\frac{3}{2}$ 0 yellow; headstitch both sides R U U R U R U U R 2 R
	Sune $\frac{3}{4}$ 1 yellow; rest clockwise R U R U R U 2 R		Chameleon $\frac{3}{2}$ 2 yellow; next to each other f R U R U r f F R F
	Anti-Sune $\frac{3}{4}$ 1 yellow; rest counter-clockwise R U 2 R U R U R		Bowtie $\frac{3}{2}$ 2 yellow diagonal f R U R U r f F R
	Pi $\frac{3}{2}$ 0 yellow; headstitch left f R U R U f F R U R U F	FALLBACK The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked; three is the worst. Top not all yellow and nothing matches? Run Sune and Read again.	

ADJACENT CORNER SWAP

exactly one face shows headlights

• T1: headlights; pairs B-left F-left
RUR'U' R'F R2UR'U' RUR'F'

Ja L solid; pairs B-right F-left R-front
x R2FRF' RU2 r'Ur U2

Jb L solid; pairs B-left F-right R-back
RUR'F' RUR'U' R'F R2UR'

Aa L headlights; pairs F-right R-front
x L2D2 L'U'L D2 L'U'L'

Ab L headlights; pairs B-right R-back
x' L2D2 LU'L D2 LU'L

F1 solid
R'U'F' RUR'U' R'F R2UR'U' RUR'U'

One face shows headlights; two corners of that face match. Hold as drawn, then turn the top to finish.

RUR'U' sexy move
 RURU sune
 R'RF'F' sledgehammer
 gap = change grip

ADJACENT CORNER SWAP

continued

Ra L headlights; pairs B-left F-right
RUR'U' RUR DR'UR'D' R'U2R'

Rb L headlights; pairs B-left F-right
R2F RURUR' F'R U2R'U2R

Ga L headlights; pairs F-right R-back
R2UR'UR'U' RUR'2 U'D R'UR'D'

Gb L headlights; pairs B-back R-front
R'UR' UD R2UR'UR'U' RUR'2D

Ge L headlights; pairs B-right R-front
R2UR'UR'U' R'UR'2 UD' R'UR'D'

Gd L headlights; pairs F-front R-back
RUR' U'D R2UR'UR'U' UR'UR'2D

Learns in the printed order, three weeks, about six weeks. Ja and Jb on the same day – one is the mirror of the other.

ADJACENT CORNER SWAP

exactly one face shows headlight

● T L headlights; pairs B-left
RUR'F' R'F R2UR'U' RUR'F'

Ja 1. solid; pairs B-right F-left front
x R2FR' RU2 r'Ur U2

Jb 1. solid; pairs B-right F-right back
RUR'F' URUR' R'F R2UR'

Aa L headlights; pairs F-right R-front
x L2D2 L'U'L D2 L'U'L

Ab L headlights; pairs B-right R-back
x' L'2D2 LU'L D2 LU'L

F1 solid
R'U'F' RUR'F' R'F R2UR'U'
RUR'UR

One face shows headlights; two corners of that face match. Hold as drawn, run, then turn the top to finish.

RUR'U sexy move **in** **RUR'U** same **o** **R'RF'** sledgehammer **gap** = change grip

ADJACENT CORNER SWAP

continued

Ra L headlights; pairs B-left
RUR'U' RUR DR'UR'D' R'U2R'

Rb L headlights; pairs B-left
R2F RURUR' F'R U2R'U2R

Ga L headlights; pairs F-right
R2UR'UR'U' RU'R2 U'D R'UR'D'

Gb L headlights; pairs B-back
R'UR' UD' R2UR'UR' RU'R2D

Ge L headlights; pairs B-right
R2UR'UR'U' RU'R2 UD' R'UR'D'

Gd L headlights; pairs B-front
RUR' U'D R2UR'UR' UR'UR'U2D'

Learns in the printed order, three a week, about six weeks; Ja and Jb on the same day – one is the mirror of the other.

ANXEN – BIG CUBES and THE MAP
□□□□□□□□

BIG CUBES

4x4 and 5x5

Reduction: build the 6 centers –> pair the 12 edges –> solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers, 2R = 2nd LAYER ONLY – never widen it.

last 2 edges

R U R' L F R' F' R

2R2 U2 2R2 U2W2

slice out, run it, slice back – both cubes

2 edge pairs swapped – half the time

2R2 U2W2

4x4 PLL parity – 1 edge/2 pair flipped – half the time

Rw U2 x Rw U2 Rw U2 Rw' U2 Rw' U2

Rw U2 x Rw U2 Rw U2 3Rw' U2 Rw U2 Rw' U2 Rw' U2

1 edge/2 pair swapped – one cube differs 5x5 has 1 PLL parity

Rw U2 Rw' U2 Rw' U2 Rw' U2

Fix parity during the last layer, before ULL and PLL – both parity algorithms move corners.

NOTATION

Each letter turns that face clockwise – so from your seat L, D and B turn COUNTER-CLOCKWISE –> counter-clockwise – 2 = half turn.

M = middle slice, turning the way L turns. x y z = z = the whole cube, turning the way R, U and F turn. Lowercase r f = that face plus the layer behind it, turning together.

BEGINNER SHORTCUT

Twist a yellow corner four turns at the top face between corners.

yellow faces RIGHT

R'D'R'D

x2

yellow faces FRONT

D'D'R'D

WORDS

algorithm = a fixed sequence of turns – trigger = a short algorithm you hands run as one unit – sexy move = the R U F U' trigger –

edge/corner = the R F F R trigger, parity = a case a 3x3 cannot you have edge pair = two edges glued into one, on a 4x4 or 5x5

RURU' sexy move **RURU** sexy **R'FFF** = a short algorithm that gain change grips

ANFNEX – BIG CUBES AND THE MAP

BIG CUBES – 4x4 and 3x3

Build the cube's 2 centers – pair the 12 edges to solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd LAYER ONLY – never widen it.

last 2 edges

4x4 PLL parity

4x4 OLL parity – 1 edge pair flipped – half the time

Rw U2 x Rw U2 Rw U2 Rw U2 slice out, run it, slice back – both cubes

Rw U2 x Rw U2 Rw U2 Rw U2 2 edge pairs swapped – half the time

Rw U2 x Rw U2 Rw U2 3Rw U2 1 edge, 3x2 has 1 PLL parity

Rw U2 x Rw U2 Rw U2 2Rw U2 2 edges, 3x2 has 1 PLL parity

Fix parity during the last layer, before OLL and PLL – both parity algorithms move corners.

NOTATION

Each letter turns that face clockwise, seen from outside the cube – so from your seat L, D and B look counter-clockwise. = counter-clockwise. 2 = half turn.

M = middle slice, turning the way L turns. x y z = the whole cube, turning the way R, U and F turn. Lowercase r f = that face plus the layer behind it, turning together.

WORDS

algebraic = a fixed sequence of turns = trigger = a short algorithm your hands run as one unit - sexy move = the R U R' U' trigger

edgeflinger = the R F R' B trigger. parity = a case a 3x3 cannot have - edge pair = 2 edges glued into one, on a 4x4 or 5x5

RURU sexy move RURU sexy RFFF short algorithm gap = change grip

BEGINNER SHORTCUT

Twist a yellow corner four turns at a time. Twist a yellow corner four turns at a time, turning ONLY the top face between corners.

yellow faces RIGHT R'D'R'D x2

yellow faces FRONT R'D'R'D x2

Print at 100% / Actual size — do NOT fit to page. Two-sided, flip on the SHORT edge. The nume

The numeral on both faces is your check: cut ONE card first. This bar must measure

■ □ □ 1/3 · B FIRST SOLVE

 FRUITS
one corner holds it FROM LEFT
PULL IT PULL

FORUM no corner home run it once from any hold, then look again

This cycles the other three corners into place. It also twists the

First turn U so at least two top edges match the center under them. The first seven moves are Sine used on every card after this one.

Yellow faces RIGHT

 KURU KURU 0

unsolved corner to front-right. Never turn the whole cube.

thing on this card that gets retired.

PLT CORNERS · highlights = two matching corners on one face · two new algorithms finish any last layer

K0K0-U- K-F K20-K-U- K0K-F- Edges: Ub. No solved edge to put at the back? Run it twice.

FRU'R'U' RUR'F' RUR'U' R'FRF'

Up the front edge travels LEFT

was reassembled wrong. Pop it out and reseat it.

tree top turn that lines a case up

7W 07W 70 7W 07W

 משרד החינוך
 מנהל מחוז תל אביב
 תל אביב

corners but wicks the yellow face, and here the yellow face is finished first.

ONE-LOOK PLL ■■■ 3/3-B

University of Würzburg

• **Up B solid**; F, L, R headlights; 3 edges clockwise
P311 P11111 P111 P1111

- **H** all 4 headlights; opposite edges swap

Nb pairs B-right F-left L-back R-front

HOW TO LOOK

7. Only then read the edges to pick the exact case.

WORDS P1 = slide the top pieces home · headlight = two matching
Beaten five times by one case? Park it, two-look it, come back next

diagonal corner swap – the swapping corners in opposite

Age Group	2007	2008	2009
18-29	~85	~85	~85
30-49	~75	~75	~75
50-69	~65	~65	~65
70+	~55	~55	~55

ANNEX – BIG CUBES AND THE MAP

down.
2/3 TWO-LOOK · +13 · unlock: fifteen last layers in exactly two
home · June = the one-yellow-corner algorithm, used on every card
after Card 1 · sexy move = the R U R' U' trigger · sledgedhammer =

After that: full OIL, F2L, cross planning, look-ahead – in the app,

diagonal corner swap = the swapping corners sit opposite

where the medium stops.

Set v1.0 · reprint any single card at cubepath-six.vercel.app/print · print at 100% / Actual size, never 'fit to page'.

אין אונזערע ארבעטן זענען דא פיל פאקטארן וואס זענען פארמאגט פאר אונזערע ארבעטן. און דאס איז דאס וואס איז פארמאגט פאר אונזערע ארבעטן.
